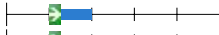
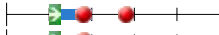
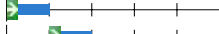
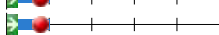
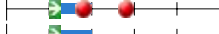
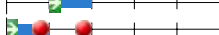
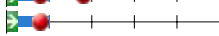


Program overview

12-Sep-2023 11:11

Year 2023/2024
Organization Architecture and the Built Environment
Education Bachelor Architecture, Urbanism and Building Sc.

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
Bachelor BK			
Bachelor 1			
BK1GR1	Fundamentals 1 House, Garden and Landscape	5	
BK1ON1	Design 1 House and Settlement in the Landscape	10	
BK1OV1	Representation, Visualization and Form 1 Space and Form	5	
BK1TE1	Technology 1 Technical Scientific Fundaments I	5	
BK1TE2	Technology 2 Materialization and Construction	5	
Bachelor 2			
BK2AC1	Academic Skills 1 Architecture as Scientific Discipline	5	
BK2GR2	Fundamentals 2 Public Space and Public Building	5	
BK2ON2	Design 2 Design and Engineering	10	
BK2OV2	Representation, Visualization and Form 2 Structure and Detail	5	
BK2TE3	Technology 3 Technical Scientific Fundaments II	5	
Bachelor 3			
BK3GR3	Fundamentals 3 Housing, Building and Surroundings	5	
BK3MA1	Society, Process and Practice 1 Planning and Program	5	
BK3ON3	Design 3 City and Public Space	10	
BK3OV3	Representation, Visualization and Form 3 Performative Design	5	
BK3TE4	Technology 4 Construction and Climate Design	5	
Bachelor 4			
BK4AC2	Academic Skills 2 Empirical Research Techniques for Architecture	5	
BK4GR4	Fundamentals 4 The European Metropolis	5	
BK4MA2	Society, Process and Practice 2 Design and Construction Management	5	
BK4ON4	Design 4 Dwelling and Dwelling Environment	10	
BK4TE5	Technology 5 Dwelling Technology	5	
Bachelor 5			
Keuze uit:			
Vrije of externe minor			
Minors BK 2023			
BK-MI-141-23 Stageminor			
BK5STA	Work in Practice: Architecture	20	
BK5STB	Work in Practice: Building Technology	20	
BK5STPO	Internship: Personal development	5	
BK5STR	Work in Practice: Management in the Built Environment	20	
BK5STU	Work in Practice: Urbanism	20	
BK5STVA	Report Work in Practice: Architecture	5	
BK5STVB	Report Work in Practice: Building Technology	5	
BK5STVR	Report Work in Practice: Management in the Built Environment	5	
BK5STVU	Report Work in Practice: Urbanism	5	
BK-MI-193-23 BK-MI-193-23 Cities Migration and Socio-Spatial Inequality (CMSI)			
BK7470	CMSI Lecture Series and Review Paper	6	
BK7471	CMSI Collaborative Project: Tackling Spatial Inequality	6	
BK7472	CMSI Engaging with Practice of Spatial Inequality	3	
BK-MI-197-23 BK-MI-197-23 Spatial Computing in Architectural Design			
BK7083	Computational Design Studio	9	
BK7084	Computational Simulations	6	
BK-MI-198-23 BK-MI-198-23 Architecture Presentation: Visions Reviewed			
BK7140	Presenting Project Facts	3	
BK7141	Presenting Project Visions	3	
BK7142	Presenting Project Prospects	9	
BK-MI-203-23 BK-MI-203-23 Archineering Q1			
BK7460-13	Archineering 1	15	
BK-MI-204-23 BK-MI-204-23 Archineering Q2			
BK7461	Archineering 2	15	
BK-MI-209-23 BK-MI-209-23 Spaces of Display Q1			
BK7064	Spaces of Display 1	10	
BK7066	Tools for Spaces of Display 1	5	

BK-MI-210-23		BK-MI-210-23 Spaces of Display Q2		
BK7065	Spaces of Display 2	10		
BK7067	Tools for Spaces of Display 2	5		
BK-MI-2016-23		BK-MI-216-23 Sustainable Urbanism: The Green-Blue City		
BK7215	Basic techniques for sustainable urban design	5		
BK7225	City and Public Space: Sustainable Urban Design	10		
BK-MI-231-23 (Re)Imagining Port Cities: Understanding Space, Society and Culture				
BK7940	(Re)imagining Port Cities: Understanding Space, Society and Culture Q1	15		
BK7941	(Re)imagining Port Cities: Understanding Space, Society and Culture Q2	15		
BK-MI-235-23 Spatial Computing for Digital Twinning				
BK7951	Spatial Decision Support Systems	5		
BK7952	3D Data Acquisition and Analysis	5		
BK7953	Open Urban Data Governance	5		
BK-MI-236-23 Timeless Typologies - "Remodelling Architecture"				
BK7942	Culture & Understanding	5		
BK7943	Design & Remodelling	10		
BK-MI-241-23 Heritage and Design				
BK7550	Landscape and Transition	5		
BK7551	History of Dutch Cities and Landscapes	5		
BK7552	Heritage: Theory and Practice	5		
BK7553	Architecture and Re-use	5		
BK7554	History of Dutch architecture and art	5		
BK7555	City and Transformation	5		
Bachelor 6				
BK6AC3	Academic Skills 3 Design Reflection	5		
BK6MA3	Society, Process and Practice 3 Management, Planning and (Re)Development	5		
BK6ON5	Design 5 Area (Re)Development in the Metropolitan Landscape	10		
BK6ON6	Design 6 Building and Technology	10		

Year	2023/2024
Organization	Architecture and the Built Environment
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Bachelor BK

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Bachelor Architecture, Urbanism and Building Sc.

Bachelor 1

BK1GR1	Fundamentals 1 House, Garden and Landscape	5
Course Coordinator	Dr. L.M. Calabrese	
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	

BK1ON1	Design 1 House and Settlement in the Landscape	10
Course Coordinator	Ir. S. Steenbruggen	
Responsible for assignments	Ir. S. Steenbruggen	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	Dutch	

BK1OV1	Representation, Visualization and Form 1 Space and Form	5
Course Coordinator	W.C. Yung	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	Ir. K.G. Kaltenbach	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Instructor	J. van der Pas	
Instructor	G.W. Bohle	
Instructor	M.G. Vink	
Responsible for assignments	P.A. Koorstra	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	

BK1TE1	Technology 1 Technical Scientific Fundaments I	5
Course Coordinator	Ir. E.R. van den Ham	
Instructor	Ir. A. Borgart	
Instructor	Ir. E.R. van den Ham	
Instructor	Ir. C.J. Janssen	
Instructor	Dr. E. Brembilla	
Responsible for assignments	Ir. E.R. van den Ham	
Co-responsible for assignments	Ir. A. Borgart	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	
Tags	Building Engineering Energy Mechanics Physics Sustainability Technology	

BK1TE2	Technology 2 Materialization and Construction	5
Course Coordinator	Ir. K.B. Mulder	
Responsible for assignments	A.H. Snijder	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	Dutch	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Bachelor Architecture, Urbanism and Building Sc.

Bachelor 2

BK2AC1	Academic Skills 1 Architecture as Scientific Discipline	5
Course Coordinator	Dr. E. Louw	
Course Coordinator	Ir. M.G.F. Overschie	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Dr. P.E. Vermaas	
Instructor	N.U. Will	
Instructor	Ir. E.J.G.C. van Dooren	
Instructor	Drs. P.C. Post	
Instructor	J.F. Michielen-van de Riet	
Responsible for assignments	Dr. E. Louw	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	Dutch	

BK2GR2	Fundamentals 2 Public Space and Public Building	5
Course Coordinator	Dr. L.M. Calabrese	
Course Coordinator	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	Dutch	

BK2ON2	Design 2 Design and Engineering	10
Course Coordinator	Ir. F.R. Schnater	
Responsible for assignments	Ir. F.R. Schnater	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	Dutch	

BK2OV2	Representation, Visualization and Form 2 Structure and Detail	5
Course Coordinator	W.C. Yung	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	Ir. K.G. Kaltenbach	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Instructor	J. van der Pas	
Instructor	G.W. Bohle	
Instructor	Ir. J.J.J.G. Hoogenboom	
Instructor	M.G. Vink	
Responsible for assignments	P.A. Koorstra	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	Dutch	

BK2TE3		Technology 3 Technical Scientific Fundaments II	5
Course Coordinator	Ir. A. Borgart		
Instructor	Ir. A. Borgart		
Instructor	Ir. E.R. van den Ham		
Instructor	S.P. de Bont		
Instructor	Ir. C.J. Janssen		
Responsible for assignments	Ir. A. Borgart		
Education Period	3		
Start Education	3		
Exam Period	3 5		
Course Language	Dutch		
Study Goals			

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Bachelor Architecture, Urbanism and Building Sc.

Bachelor 3

BK3GR3	Fundamentals 3 Housing, Building and Surroundings	5
Course Coordinator	Ir. W.W.L.M. Wilms Floet	
Course Coordinator	Dr.ir. N.M.J.D. Tillie	
Responsible for assignments	Ir. W.W.L.M. Wilms Floet	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	Dutch	

BK3MA1	Society, Process and Practice 1 Planning and Program	5
Course Coordinator	Dr. H.J.F.M. Boumeester	
Course Coordinator	Prof.dr.ir. M.G. Elsinga	
Responsible for assignments	Dr. H.J.F.M. Boumeester	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	

BK3ON3	Design 3 City and Public Space	10
Course Coordinator	Ir. E.M. Bet	
Course Coordinator	Ir. T. Bouma	
Responsible for assignments	Ir. E.M. Bet	
Co-responsible for assignments	Ir. T. Bouma	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	Dutch	
Period of Education	Quarter	

BK3OV3	Representation, Visualization and Form 3 Performative Design Computation	5
Course Coordinator	P. de Ruiter	
Instructor	Prof.dr.ir. I.S. Sariyildiz	
Instructor	M. Turrin	
Instructor	Dr. S. Asut	
Instructor	Ir. J.J.J.G. Hoogenboom	
Responsible for assignments	P. de Ruiter	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	

BK3TE4	Technology 4 Construction and Climate Design	5
Course Coordinator	Ir. C.J. Janssen	
Responsible for assignments	Ir. C.J. Janssen	
Responsible for assignments	Dr. S. Brancart	
Responsible for assignments	M.M.E. van Esch	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	
Expected prior knowledge	The minimum required levels of knowledge to pass the TE1, TE2 and TE3 course are a prerequisite for the BK3-TE4.	
Course Contents	The student examines, for an existing building, how design of construction, structure and M&E systems relate to each other. In addition the student examines the influence of the building on the urban context and the influence of the urban context on the building. Design of construction, structure, M&E installations and urban technology each have a specific role which are designed in relation to each other.	
Study Goals	De student is able to: 1. recognise and describe different types of building constructions and determine dimensions and details of the different parts, [BC] 2. recognise and describe different types of structure and determine dimensions of different parts using relevant norms, [DC] 3. recognise and describe different M&E systems and evaluate interior climate on energy efficiency (heating, cooling, ventilation), [KO] 4. recognise and describe the influence of technical urban factors (sun, wind, water and vegetation) on the urban microclimate. 5. deze kennis toe te passen bij de integrale analyse van een gemiddeld complex gebouw en de openbare ruimte, [BC+DC+KO+ST] 5. apply this knowledge on an integral analyses of a relatively complex building it's urban context, [BC+DC+KO+ST] 6. explain in text, drawings and/or calculations how a relatively complex building and the surrounding urban context can be modified to improve sustainability, circularity and climate adaptivity. [BC+DC+KO+ST]	
Education Method	The module has two types of education: lectures in which the theoretical knowledge is explained and a practical assignment in which, for existing buildings, the systems for M&E, structure, and design of construction are analysed and evaluated. In addition a relation to the urban context of the building is made. During the practical assignment different sub-teams analyse a building and its surroundings. Attendance to the tutoring sessions of the practical assignment is obligatory. By failing to attend the first tutoring session the student loses the right to attend to the remaining sessions. The result of the analyses and evaluation will be presented and documented in a report.	
Literature and Study Materials	See resources and tools / literature on Brightspace	
Assessment	The student will receive one final grade for the required submissions. The final grade is determined by taking the average of two partial grades: one for the exam (50%) and one for the report + presentation (50%). In the exam theoretical knowledge about design of construction, structure, M&E systems and Urban technology is tested. In the practical assignment an existing building is analysed. This analysis will be graded by the quality of the report and the presentation. The quality of the report and presentation are based on completeness, clarity, and the quality / correctness of the individual products.	
Special Information	After submission, grades are returned after a maximum of 15 working days.	
Period of Education	Half a quarter	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Bachelor Architecture, Urbanism and Building Sc.

Bachelor 4

BK4AC2	Academic Skills 2 Empirical Research Techniques for Architecture	5
Course Coordinator	Dr. H.J.F.M. Boumeester	
Course Coordinator	Dr. M.E.A. Haffner	
Instructor	Dr. H.J.F.M. Boumeester	
Instructor	Dr. M.E.A. Haffner	
Instructor	Dr. H.M.H. van der Heijden	
Instructor	Dr. J.S.C.M. Hoekstra	
Instructor	Dr. E. Louw	
Instructor	Dr.ir. S. Zijlstra	
Instructor	Dr. S.J.T. Jansen	
Instructor	A. Petrovi	
Instructor	T.E. Jylhä	
Instructor	Ir. E.S. Meerburg	
Instructor	R. van der Drift	
Instructor	Drs. W.H. Hoffmans	
Responsible for assignments	Dr. H.J.F.M. Boumeester	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	Dutch	

BK4GR4	Fundamentals 4 The European Metropolis	5
Course Coordinator	Ir. W.W.L.M. Wilms Floet	
Course Coordinator	Dr.ir. N.M.J.D. Tillie	
Responsible for assignments	Ir. W.W.L.M. Wilms Floet	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	Dutch	

BK4MA2	Society, Process and Practice 2 Design and Construction Management	5
Course Coordinator	Ing. R.J.G. van Warmerdam	
Course Coordinator	Dr. H.M.H. van der Heijden	
Responsible for assignments	Dr. H.M.H. van der Heijden	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	Dutch	

BK4ON4	Design 4 Dwelling and Dwelling Environment	10
Course Coordinator	Ir. C.M. Calis	
Responsible for assignments	Ir. C.M. Calis	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	Dutch	

BK4TE5	Technology 5 Dwelling Technology	5
Course Coordinator	Ir. K.B. Mulder	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	Dutch	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Bachelor Architecture, Urbanism and Building Sc.

Bachelor 5

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Bachelor Architecture, Urbanism and Building Sc.

Keuze uit:

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Bachelor Architecture, Urbanism and Building Sc.

Vrije of externe minor

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Bachelor Architecture, Urbanism and Building Sc.

Minors BK 2023

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Bachelor Architecture, Urbanism and Building Sc.

BK-MI-141-23 Stageminor

BK5STA	Work in Practice: Architecture	20
Course Coordinator	Dr. O. Caso	
Responsible for assignments	Dr. O. Caso	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	Dutch	

BK5STB	Work in Practice: Building Technology	20
Course Coordinator	Ir. F.R. Schnater	
Responsible for assignments	Ir. F.R. Schnater	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	Dutch	
Course Contents	this minor is in Dutch only	

BK5STPO	Internship: Personal development	5
Course Coordinator	Dr. J.L. Heintz	
Instructor	Prof.dr. P.W.C. Chan	
Responsible for assignments	Dr. J.L. Heintz	
Education Period	1 3 Different, to be announced	
Start Education	1 3	
Exam Period	none	
Course Language	Dutch	

BK5STR	Work in Practice: Management in the Built Environment	20
Course Coordinator	K.B.J. Van den Berghe	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	Dutch	
Special Information	The maximum marking period is 10 work days.	

BK5STU	Work in Practice: Urbanism	20
Course Coordinator	R.J. van der Veen	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	Dutch	

BK5STVA	Report Work in Practice: Architecture	5
Course Coordinator	Dr. O. Caso	
Responsible for assignments	Dr. O. Caso	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	Dutch	

BK5STVB	Report Work in Practice: Building Technology	5
Course Coordinator	Ir. F.R. Schnater	
Responsible for assignments	Ir. F.R. Schnater	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	Dutch	

BK5STVR	Report Work in Practice, Management in the Built Environment	5
Course Coordinator	Ir. C.G. Luth	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	Dutch	

BK5STVU	Report Work in Practice: Urbanism	5
Course Coordinator	R.J. van der Veen	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	Dutch	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Bachelor Architecture, Urbanism and Building Sc.

BK-MI-193-23

BK7470	CMSI Lecture Series and Review Paper	6
Course Coordinator	Dr. R.J. Kleinhans	
Instructor	Dr.ir. G. Slingerland	
Instructor	Dr. T. Verbeek	
Instructor	D. Adlakha	
Responsible for assignments	Dr. R.J. Kleinhans	
Contact Hours / Week x/x/x/x	6 hours per week, starting from week 1.1 and ending in week 1.9.	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	This first course of the minor Cities, Migration and Socio-Spatial Inequality (CMSI) offers interactive lectures introducing the context, themes and key concepts in the minor: socio-spatial inequality, spatial justice, migration, diversity, and resilience. For the interactive lectures, background literature will be provided on Brightspace. In turn, students will prepare statements and questions in relation to each lecture. Some of the lectures will be provided by instructors from other universities. After a few weeks, students will select a theme or concept based on the lectures and write a literature review paper (max. 4200 words) with a research question that is approved by the instructors. This review should not only include a discussion of one of the key concepts, but also an analysis of its spatial implications. During the paper writing process, regular feedback will be provided by instructors. Each student will be assigned to a paper tutor for individual feedback (with a maximum of five students per instructor in order to enable rapid responses).	
Course Contents Continuation	This course is part of the 15 ECTS CMSI minor intended for students who are highly motivated to develop an interdisciplinary perspective on socio-spatial inequality, migration, diversity and resilience. Students from different backgrounds will acquire a thorough understanding of these concepts and learn how to analyse related questions from the academic disciplines of urban geography, sociology, urban planning and design, and public administration. Students will also be trained to apply this knowledge in plans with socio-spatial strategies for intervention in and redesign of concrete planning cases. CMSI Minor participants must have a passion for social scientific research and combine a strong academic curiosity with a determination to apply interdisciplinary knowledge in real-life situations of complex urban planning cases in the Netherlands.	
Study Goals	After this course, the students are able to: - Define the key concepts socio-spatial inequality, segregation, spatial justice, resilience, and diversity; - Discuss the meaning of these concepts in the context of their theoretical background and the broader societal context; - Identify how socio-spatial inequality, migration and diversity manifest themselves on the level of cities and neighbourhoods, and provide examples of this manifestation; - Synthesize the relevant literature for a chosen theme or concept and analyse its spatial implications.	
Education Method	Interactive lectures, in-class discussions, literature study and writing a review paper. Students will also deliver (in pairs) short pitches, called 'Your Take on Social Inequality'. In these pitches, students choose and explain specific news items or topics related to the overall minor theme, and spark debate in class.	
Literature and Study Materials	Background literature will be provided on Brightspace. If a book needs to be bought, this will be indicated on Brightspace before the actual start of the course.	
Practical Guide	A syllabus of the three CMSI minor courses will be provided on Brightspace	
Assessment	Instructor feedback on the literature review paper will be provided in writing and face-to-face at pre-scheduled moments during the writing process. The summative assessment is the actual grading of the paper. Students should achieve a minimum grade of 5.5 for the paper. If students fail to achieve the pass mark of 5.5, they have an opportunity to resubmit an improved version of the literature review paper within four weeks from the last work day of quarter 1. In this case, the maximum score will be sufficient, meaning that the student passes this course.	
Tags	Building & Spatial Development Policy Analysis Sustainability	
Period of Education	Quarter 1	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

BK7471	CMSI Collaborative Project: Tackling Spatial Inequality	6
Course Coordinator	Dr. R.J. Kleinhans	
Instructor	Dr. R.J. Kleinhans	
Instructor	Dr.ir. G. Slingerland	
Instructor	Dr. T. Verbeek	
Instructor	D. Adlakha	
Responsible for assignments	Dr. R.J. Kleinhans	
Contact Hours / Week x/x/x/x	5 hours per week, starting from week 1.2 and ending in week 1.10.	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	This second course of the minor 'Cities, Migration and Socio-Spatial Inequality' consists of interdisciplinary group work, in which students from different backgrounds work together on a real-life case study with a complex set of problems related to inequality, diversity and migration, in an urban setting. The case study will build on the action research course problem analysis and stakeholder identification (within BK7472) and planning and design, i.e. creating an integrated socio-spatial strategy that addresses the identified problems and opportunities. Obviously, students are encouraged to align their literature review papers (BK7470) with the research part of the project. The group work extends over almost the full length of the minor and starts with forming a project team of approximately three students. Each team will draft a collaborative project report, outlining the chosen topic, research and design approach. Towards the end of the minor, the teams will present their case study analysis and design, process the feedback from instructors and fellow students, and subsequently submit the final version of the collaborative project report.	
Course Contents Continuation	This course is part of the 15 ECTS CMSI minor intended for students who are highly motivated to develop an interdisciplinary perspective on socio-spatial inequality, segregation, migration, diversity and resilience. Students from different backgrounds will acquire a thorough understanding of these concepts and learn how to analyse related questions from the academic disciplines of urban geography, sociology, urban planning and design, and public administration. Students will also be trained to apply this knowledge in plans with socio-spatial strategies for intervention in and redesign of concrete planning cases. CMSI Minor participants must have a passion for social scientific research and combine a strong academic curiosity with a determination to apply interdisciplinary knowledge in real-life situations of complex urban planning cases in the Netherlands.	
Study Goals	After this course, the students are able to: - Analyse the nature of socio-spatial inequality, diversity, migration, and resilience and their effects on various spatial (mostly local) levels and from an interdisciplinary perspective; - Assess the planning, governance and design implications of socio-spatial inequality, migration and diversity at the urban and neighbourhood level; - Develop plans with socio-spatial strategies for intervention.	
Education Method	Students will be assigned to interdisciplinary project teams of three students by the course co-ordinator. Each team will draft a project plan, outlining the chosen topic, research and design approach. Instructors who will provide coaching and feedback to the students in consecutive meetings. Towards the end of the minor, the teams will present their case study analysis and design, process the feedback from instructors and fellow students, and subsequently submit the final version of the collaborative project report. Active participation and attendance in the course sessions (two per week) is compulsory. Repeated absence without notifying the course co-ordinator in advance will result in deduction of points for the individual student (not the group).	
Literature and Study Materials	Useful literature and other information will be provided on Brightspace.	
Practical Guide	A syllabus of the three CMSI minor courses will be provided on Brightspace.	
Assessment	Instructor feedback on the research and design process and outcomes will be provided at scheduled moments during the project. The summative assessment is the actual grading of the report. Students should achieve a minimum grade of 5.5 for the collaborative project report. If students fail to achieve the pass mark of 5.5, they have an opportunity to resubmit an improved version of the collaborative project report within four weeks from the last work day of quarter 1. In this case, the maximum score will be sufficient, meaning that the student passes this course.	
Tags	Building & Spatial Development Fieldwork Policy Analysis Sustainability	
Period of Education	Quarter 1	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

BK7472	CMSI Engaging with Practice of Spatial Inequality	3
Course Coordinator	Dr. T. Verbeek	
Instructor	Dr. R.J. Kleinhans	
Instructor	Dr.ir. G. Slingerland	
Instructor	Dr. T. Verbeek	
Instructor	D. Adlakha	
Responsible for assignments	Dr. R.J. Kleinhans	
Contact Hours / Week x/x/x/x	4 hours per week, starting from week 1.1 and ending in week 1.5.	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	This third course in the minor Cities, Migration and Socio-Spatial Inequality (CMSI) consists of an action research approach in a neighbourhood in Rotterdam or another city. Action research is a research method that aims to simultaneously investigate real-life problems and provide practical solutions, usually in a collaborative way together with practitioners and community members. Using a range of research techniques (observations, street interviews, in-depth interviews, secondary data analysis), students will collect data that enable an integrated, multidisciplinary analysis of a neighbourhood, focussing on socio-spatial inequality, diversity and social resilience (in Dutch: veerkracht). The multidisciplinary analysis will uncover socio-spatial inequality and diversity across several domains, ranging from housing and public space to education, employment, accessibility, participation and safety. The collected data and findings will be synthesised and visualised in an action research report, which forms the basis of the collaborative strategic plans that will be developed in the BK7471 course.	
Course Contents Continuation	This course is part of the 15 ECTS CMSI minor intended for students who are highly motivated to develop an interdisciplinary perspective on socio-spatial inequality, segregation, migration, diversity and resilience. Students from different backgrounds will acquire a thorough understanding of these concepts and learn how to analyse related questions from the academic disciplines of urban geography, sociology, urban planning and design, and public administration. Students will also be trained to apply this knowledge in plans with socio-spatial strategies for intervention and redesign in concrete cases.	
Study Goals	CMSI Minor participants must have a passion for social scientific research and combine a strong academic curiosity with a determination to apply interdisciplinary knowledge in real-life situations of complex urban challenges in the Netherlands. After this course, students are able to: - Apply different action research techniques in a critical and rigorous way to a social inequality challenge in a diverse urban neighbourhood; - Evaluate various data and findings in an integrated multidisciplinary analysis, considering different perspectives; - Present collected data, findings, and interpretations in a convincing way, using appropriate visualization.	
Education Method	The course provides instruction and training in different research techniques, in particular observations, street interviews, in-depth interviews and desk research (secondary data analysis). There will also be attention to attractive visualization and correct reporting of findings. The training offers students a kick-start for their on-site research and data collection that will feed into their strategic plan (BK7471). Throughout the course, supervision and regular feedback will be provided by the course instructors. Active participation and attendance in the course sessions (one per week) is compulsory. Repeated absence without notifying the course co-ordinator in advance will result in deduction of points for the individual student (not the group).	
Literature and Study Materials	Useful literature and other information will be provided on Brightspace.	
Practical Guide	A syllabus of the three CMSI minor courses will be provided on Brightspace.	
Assessment	Instructor feedback on the research process and findings will be provided on a weekly basis. The summative assessment is the actual grading of the report. Students should achieve a minimum grade of 5.5 for the action research report. If students fail to achieve the pass mark of 5.5, they have an opportunity to resubmit an improved version of the action research report within four weeks from the last work day of quarter 1. In this case, the maximum score will be sufficient, meaning that the student passes this course.	
Tags	Analysis Building & Spatial Development Fieldwork Group work Integrated Policy Analysis Project Research Methods Sustainability	
Period of Education	Quarter 1	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Bachelor Architecture, Urbanism and Building Sc.

BK-MI-197-23

BK7083	Computational Design Studio	9
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Course Coordinator	Prof.dr.ir. I.S. Sariyildiz	
Instructor	Ir. J.J.J.G. Hoogenboom	
Responsible for assignments	Prof.dr.ir. I.S. Sariyildiz	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	The objective of the course is to design a multi-functional building complex, based on preconditions and environmental data, by utilizing computational tools, methods, and techniques in a workflow. The students will design basic algorithms to measure and optimize the performance of the building concerning the urban, environmental, and sustainability features. The course will explore how computational design and -thinking can enhance architectural design and its processes. The computational design models will be set up systematically and flexibly using (visual) programming tools. The Computational Design Studio runs alongside Computational Simulations (BK7084) and both form the Minor Spatial Computing in Architectural Design (BK-MI-197). This minor is the second part of the Minor Spatial Computing for Sustainable Development. For more information about the complete minor program check: https://www.tudelft.nl/bk/studeren/minoren-en-keuzevakken/spatial-computing-for-sustainable-development.	
Study Goals	By the end of this course the students should be able to: - collect and use data for decision support for the design process (data-driven design), - develop computational tools to measure and optimize performance, - use scripting or programming language to automate (part of) the design process, - set up a rule-based system to computationally generate a design, based on a set of design requirements, - present the design and the design methods coherently with the use of XR technologies.	
Education Method	Lectures, workshops, consultations, design studio (self-study).	
Assessment	Computer Use: A laptop is required (TU Delft student laptop with dedicated GPU) he assessment is based on the deliverables as specified in the course syllabus. Design examination for group work and individual contribution.	
Tags	Algorithmics Building & Spatial Development Design Programming	
Period of Education	Quarter	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

BK7084	Computational Simulations	6
Course Coordinator	Prof.dr. E. Eisemann	
Course Coordinator	Dr. R. Marroquim	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basic mathematical skills (vector calculus, linear algebra).	
Course Contents	This course investigates the use of simulation and optimization algorithms, with a focus on image synthesis and light transport, which can be of use in the context of interior and architectural design, as well as urban planning.	
Study Goals	The students will be able to apply aspects of physically-based light simulation, transformations and procedural content generation to architectural design and urban planning. They will learn how to code simple prototypes themselves to verify their knowledge.	
Education Method	The course is organized in lectures and practicals. Further, there are exercise and instruction sessions. The content of the lectures will be partly reproduced in the practicals and a final project will integrate different elements of the practicals.	
Computer Use	The course will contain a programming component. Coding skills can be acquired during the course, which will include introductory lectures.	
Assessment	The course is evaluated by two means: an exam (can contain open and multiple-choice questions) and a final project (in small groups). The exam accounts for 30% of the final grade, and the final project accounts for 70%. To pass the course, the minimum grade for the exam and project is 5, and the minimum final grade is 6 (after rounding). There is an exam retake (might take the form of an oral exam). There is no project retake or repair.	
Period of Education	Quarter 2	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Bachelor Architecture, Urbanism and Building Sc.

BK-MI-198-23

BK7140	Presenting Project Facts	3
Course Coordinator	M.G. Vink	
Course Coordinator	P.A. Koorstra	
Instructor	W.C. Yung	
Instructor	M.G. Vink	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	12 hours a week for two weeks	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor 1-4	
Course Contents	The minor Architectural Presentations, BK-MI-198 consists of three successive courses for a total of 15 ECTS. Throughout these courses students explore representation techniques in 2D, 3D and 4D, to communicate architectural projects as they are lived in, moved through and experienced. In this first part of the minor, students will choose a case study from a number of given architectural precedents. They will work towards a high level of project understanding and present their insights and interpretations by means of a recapitulatory 2D drawing accompanied by an oral presentation. The students explore the drawing as a product telling a story. By stitching together multiple types of drawing in a total composition, students are able to show different insights into a case study project. The assignment consists of a study of example drawings, the re-drawing of a case study and the composition of a final drawing using the studied techniques.	
Study Goals	After finishing this course, the student is able to: - use various 2D drawing techniques to effectively represent and communicate a chosen case study project into a scenic drawing - develop new insights and presentations concerning the further expression and re-interpretation of the case study project, - develop creative and technical skills in the effective representation and communication of a chosen case project, using various representation techniques. - the ability to work critically, carrying out analyses, documenting results, visualizing proposals and bringing these across to a professional audience. Academic competences - Synthesis of project facts, character, ideas and ideals. - Transformation of the case project to a contemporary project presentation.	
Education Method	Course themes will be introduced in the syllabus and lectures. Consults will be held in small groups and individually - depending on the assignment. There will be two contact moments per week. We will make use of BrightSpace and personal online blogs to submit share and comment on results. Assignments are executed in a design studio setting. On Tuesdays and Thursdays students are expected to be in the studio. For the other days it is recommended to work in the studio, but it is up to the student to decide what works best. Students will keep a personal blog/site to present own results, group results and reflections on their progress. The blog needs to be updated at the end of each week.	
Assessment	The minor consists of a series of integrated exercises which are all assessed separately. All these assignments are assessed by the requested presentation products and guided by an oral presentation. These are consequently: - 2D drawing (BK7140) - 3D model (BK7141) - 4D film (BK7142) - Pecha Kucha slide show presentation (BK7142) The tutors make the assessment based on: - completeness of requested products - readability of the representation products - quality of the representation products - creativity and coherence in the representation products - relevance of the produced products in relation to the studied project - coherence between oral presentation and representation products The marks for the various exercises are translated to the grades per course as follows: - BK7140 - 3 ECTS : 2D drawing - BK7141 - 3 ECTS : 3D model - BK7142 - 9 ECTS : 4D moving image + Pecha Kucha slide show presentation	
Special Information	Coordinator	
Period of Education	Q2 of the first semester.	
Leerstoel	Formstudies, Architecture	
Minimum number of participants	15	
Maximum number of participants	25	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

BK7141	Presenting Project Visions	3
Course Coordinator	M.G. Vink	
Course Coordinator	P.A. Koorstra	
Instructor	W.C. Yung	
Instructor	M.G. Vink	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	12 hours a week for two weeks	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor 1-4	
Course Contents	The minor Architectural Presentations, BK-MI-198 consists of three successive courses for a total of 15 ECTS. Throughout these courses students explore representation techniques in 2D, 3D and 4D, to communicate architectural projects as they are lived in, moved through and experienced. Continuing on their chosen case study project from the first part of the minor, in this second part of the minor the students shift towards a 3D representation of the project. Students will focus on the material and evocative qualities of the physical model. The model is approached as a semi-independent character rather than a literal miniature building. Working on scale, students in small groups are translating the case studies into models that are both abstract and recognizable with the aim to communicate the character of the design idea.	
Study Goals	After finishing this course, the student is able to: - understand the expressive qualities of physical scale models in relation to scale - develop new insights and presentations concerning the further expression and re-interpretation of the case study project, - develop creative and technical skills in the effective representation and communication of a chosen case project, using various representation techniques. - the ability to work critically, carrying out analyses, documenting results, visualizing proposals and bringing these across to a professional audience.	
Education Method	Course themes will be introduced in the syllabus and lectures. Consults will be held in small groups and individually - depending on the assignment. There will be two contact moments per week. We will make use of BrightSpace to communicate announcements and for students to upload assignments. Assignments are executed in a design studio setting. On the scheduled days students are expected to be in the studio. For the other days it is recommended to work in the studio, but it is up to the student to decide what works best.	
Assessment	The minor consists of a series of integrated exercises which are all assessed separately. All these assignments are assessed by the requested presentation products and guided by an oral presentation. These are consequently: - 2D drawing (BK7140) - 3D model (BK7141) - 4D film (BK7142) - Pecha Kucha slide show presentation (BK7142) The tutors make the assessment based on: - completeness of requested products - readability of the representation products - quality of the representation products - creativity and coherence in the representation products - relevance of the produced products in relation to the studied project - coherence between oral presentation and representation products The marks for the various exercises are translated to the grades per course as follows: - BK7140 - 3 ECTS : 2D drawing - BK7141 - 3 ECTS : 3D model - BK7142 - 9 ECTS : 4D moving image + Pecha Kucha slide show presentation	
Period of Education	Q2	
Leerstoel	Formstudies, Architecture	
Minimum number of participants	15	
Maximum number of participants	25	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

BK7142	Presenting Project Prospects	9
Course Coordinator	M.G. Vink	
Course Coordinator	P.A. Koorstra	
Instructor	W.C. Yung	
Instructor	M.G. Vink	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	12 hours a week for two weeks	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor 1-4	
Course Contents	The minor Architectural Presentations, BK-MI-198 consists of three successive courses for a total of 15 ECTS. Throughout these courses students explore representation techniques in 2D, 3D and 4D, to communicate architectural projects as they are lived in, moved through and experienced. In the final and largest part of the minor time and thus, movement and change are added as a layer of presentation in the production of a moving image. Using and transforming the drawings and models, the production of a short film forces students to think of the case study projects in architectural qualities, sequence and atmosphere. Students will translate this into their personal cinematographic vision which will be leading in the development of their own individual film. The final screening of the film will be accompanied with an oral slideshow presentation explaining the choices made during the process of making the short film.	
Study Goals	After finishing this course, the student is able to: - create a moving image on the basis of self-produced physical scale models and/or scenic drawing (part I and II products) which represents a clear understanding of a given architectural case study project - develop new insights and presentations concerning the further expression and re-interpretation of the case study project, develop creative and technical skills in the effective representation and communication of a chosen case project, using various representation techniques. - the ability to work critically, carrying out analyses, documenting results, visualizing proposals and bringing these across to a professional audience. Academic competences - Synthesis of project facts, character, ideas and ideals. - Transformation of the case project to a contemporary project presentation. Course themes will be introduced in the syllabus and lectures.	
Education Method	Consults will be held in small groups and individually - depending on the assignment. There will be two contact moments per week. We will make use of BrightSpace to communicate announcements and for students to upload assignments. Assignments are executed in a design studio setting. On the scheduled days students are expected to be in the studio. For the other days it is recommended to work in the studio, but it is up to the student to decide what works best.	
Assessment	The minor consists of a series of integrated exercises which are all assessed separately. All these assignments are assessed by the requested presentation products and guided by an oral presentation. These are consequently: - 2D drawing (BK7140) - 3D model (BK7141) - 4D film (BK7142) - Pecha Kucha slide show presentation (BK7142) The tutors make the assessment based on: - completeness of requested products - readability of the representation products - quality of the representation products - creativity and coherence in the representation products - relevance of the produced products in relation to the studied project - coherence between oral presentation and representation products The marks for the various exercises are translated to the grades per course as follows: - BK7140 - 3 ECTS : 2D drawing - BK7141 - 3 ECTS : 3D model - BK7142 - 9 ECTS : 4D moving image + Pecha Kucha slide show presentation	
Period of Education	Q2	
Leerstoel	Formstudies, Architecture	
Minimum number of participants	15	
Maximum number of participants	25	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Bachelor Architecture, Urbanism and Building Sc.

BK-MI-203-23

BK7460-13	Archineering 1	15
Course Coordinator	Ir. S.H. Verkuijlen	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Parts	Circular Building industry, Bio-based materials, Design Process, Experimenting, Digital Fabrication, Materialization & Detailing	
Course Contents	In the minor Archineering you will learn about Circulair Building Industry, where harvesting demolition waste provides the main resource for new buildings, and Bio-Based Materials which can be harvested in a sustainable way. At the same time you will improve your skills as a designer. With the help of a team of experts (crafts(wo-)men, architects, structural and climate engineers) You will learn about materials and techniques by "building it yourself". Parallel to this, a variety of speakers is invited to give lectures on circularity, energy use and construction to explain the bigger picture. More experienced design-students will be able to improve their design skills on specific aspects like Bio-Based Materials, Timber Construction and Digital Fabrication. Students who are new in the field of architecture will explore and discover the fun of learning by designing, making and breaking.	
Study Goals	In the minor Archineering you will learn to - understand how circular design and the use of bio-based materials can help reduce climate change. - design and work with different bio-based materials (wood, hempcrete etc.). - understand woodworking techniques (by working with wood in a workshop setting). - optimise a wooden structure by building and testing. - translate a design task into a complete and working draft proposal within 2 weeks. - explore construction, climate design, digital manufacturing, detailing, and be able to experiment with these design aspects in your own design project. - build, test analyse and improve several prototypes and models on different information and scale levels.	
Education Method	Design Projects: In Archineering 1 students are gradually introduced to circular design and using bio-based materials, by means of several seminars and a series of small, two or three week design exercises often with a given topic and guiding theme. The integration of materialisation and architecture is examined by focusing on one single aspect in every design exercise, for example: Wood as a construction material, detailing with bio-based materials or climate design. Excursions and Workshops: Each design exercise starts with an excursion, lectures or workshop on the particular technical aspect that is focused on (construction/environment design/detailing). These make the students playfully familiar with the specific technical challenges within the work field of the central theme. After this, the student will practice and experiment with the knowledge gained in his own design project. Reflection: After the design assignments a reflection report is written and handed in.	
Prerequisites	This course is open to any Bachelor student who wants to learn about, how Circular Design and the use of Bio-based materials can help reduce the impact of our Building Industry on our Climate and improve their design skills at the same time. Students from every faculty at TU Delft should have received the adequate knowledge prior to entering this minor. Also, students from other universities or in HBO education can apply for the minor. In this case an email to the course coordinator is required to explain why the student wants to follow this course. This will prevent misunderstandings about the course content. This minor consists of 2 quarters (both of 15 ECs); each with separate content. For students of the Faculty of Architecture and the Built Environment TU Delft it is possible to follow either just quarter 1 or just quarter 2. For all other students who are admissible to this minor, it is only possible to follow both quarter 1 and 2 or just quarter 1 (only quarter 2 is not possible). However, it is advised to follow both quarter 1 and 2. Please be aware that each quarter has a separate minorcode. To follow both quarter 1 and 2, you should register for both minor codes in Osiris!	
Assessment	Design proposal * = Verbal presentation, (personal or as a team) with posters, on-screen presentation and lots of sketch models, prototypes that were built during the course. Reflection Paper ** = Analyses in text, photos and diagrams. * > We do quite a lot of design projects, even compared to other courses at The Faculty of Architecture and the Built Environment. This is to let the students really experience as much as possible the different aspects of the design process. By training designing, engineering and prototyping in a number of short exercises and explicitly studying and reflecting on the design process, more insight and understanding (in the personal design process and integration of technology & design) will be acquired. ** > students do an overall reflection at the end of Archineering 1.	
Period of Education	1 Quarter	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Bachelor Architecture, Urbanism and Building Sc.

BK-MI-204-23

BK7461	Archineering 2	15
Course Coordinator	Ir. S.H. Verkuijlen	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Parts	Circular Building industry, Bio-based materials, Design Process, Experimenting, Digital Fabrication, Materialization & Detailing	
Course Contents	In the minor Archineering you will learn about Circulair Building Industry, where harvesting demolition waste provides the main resource for new buildings, and Bio-Based Materials which can be harvested in a sustainable way. At the same time you will improve your skills as a designer. With the help of a team of experts (crafts(wo-)men, architects, structural and climate engineers) You will learn about materials and techniques by "building it yourself". Parallel to this, a variety of speakers is invited to give lectures on circularity, energy use and construction to explain the bigger picture. More experienced design-students will be able to improve their design skills on specific aspects like Bio-Based Materials, Timber Construction and Digital Fabrication. Students who are new in the field of architecture will explore and discover the fun of learning by designing, making and breaking.	
Study Goals	In the minor Archineering you will learn to - understand how circular design and the use of bio-based materials can help reduce climate change. - design and work with different bio-based materials (wood, hempcrete etc.). - understand woodworking techniques (by working with wood in a workshop setting). - optimise a wooden structure by building and testing. - translate a design task into a complete and working draft proposal within 2 weeks. - explore construction, climate design, digital manufacturing, detailing, and be able to experiment with these design aspects in your own design project. - build, test analyse and improve several prototypes and models on different information and scale levels.	
Education Method	Design Project: In Archineering 2 the knowledge and design methodology gained in Archineering 1 is taken into to practice by working on one integrated design project with greater complexity. The aim is to design and realise, in a team, a large scale (if possible full scale) prototype or project. Parallel to this, a variety of speakers is invited to give lectures on circularity, energy use and construction in relation to the project that will be built. Reflection: After the design assignment a reflection report is written and handed in.	
Prerequisites	The prior knowledge from Archineering 1 is strongly advised for taking part in Archineering 2. This minor consists of 2 quarters (both of 15 ECs); each with separate content. For students of the Faculty of Architecture and the Built Environment TU Delft it is possible to follow either just quarter 1 or just quarter 2. however, it is advised to follow both Archineering 1 and Archineering 2. For all other students it is not possible to only follow quarter 2 without having done quarter 1 because of certain design skills that are discussed in Archineering 1. Please be aware that each quarter has a separate minorcode. To follow both Archineering 1 and 2, you should register for both minor codes in Osiris!	
Assessment	Design proposal = Verbal presentation, (as a team) with posters, on-screen presentation and lots of sketch models, prototypes that were built during the course. Reflection Paper = Analyses in text, photos and diagrams. Students are asked to keep a personal log book that will be part of the reflection paper.	
Period of Education	1 Quarter	
Leerstoel	The minor Archineering is a coherent package of courses in the field of Architectural Engineering (aE). It is strongly related to the aE-MSc 1 course Buckey Lab, the aE-MSc 2 course van Gezel tot Meester and the aE- MSc 3+4 Graduation Studio Intecture.	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Bachelor Architecture, Urbanism and Building Sc.

BK-MI-209-23

BK7064	Spaces of Display 1	10
Course Coordinator	Ir. L.M.M. de Wit	
Instructor	Ir. S. Pietsch	
Responsible for assignments	Ir. L.M.M. de Wit	
Contact Hours / Week x/x/x/x	X/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Required for	Bachelor students Architecture and Industrial Design (TU Delft and Eindhoven) Extern HBO: Design Academy Eindhoven Art academies; graduates in Architectural Design, interior architecture	
Expected prior knowledge	Bachelor 1 to 4 or equivalent. or: Three years of HBO or advanced Bachelor in abovementioned educations. Recommended skills: Revit, Rhino, Autocad, Illustrator, Sketchup or equivalent.	
Summary	The Chair of Interiors Buildings Cities is concerned with making buildings, in places, for people. It conceives of the city, at each scale, as a work of architecture and, hence, the responsibility of the architect. This developing discussion of the chair is contextualized through an annual theme. During this interiors minor, students design a (small) exhibition space (Q1) and a fully detailed fragment of the building or the interior (Q2). We emphasize on the use of scale models as a means of research. This minor consists of 2 quarters (both of 15 ECs); each with separate content. For students of the Faculty of Architecture and the Built Environment TU Delft it is possible to follow either just quarter 1 or just quarter 2, or to follow both quarter 1 and 2. For all other students who are admissible to this minor, it is only possible to follow either quarter 1, or both quarter 1 and 2. Please be aware that each quarter of this minor has a separate minor code. If you would like to follow both quarter 1 and 2, you should register for both minor codes in Osiris.	
Course Contents	Q1 BK-MI-209 During the first quarter, students design a small exhibition space for a carefully selected artefact and its users at a specific location in a city (BK7064 Spaces of Display 1). The Tools Course (BK7066 Tools for Spaces of Display 1) focuses on specific skills that contribute to a coherent design project. Students learn how to work within the complex combination of these 'tools'. Various workshops and lectures focus on the position of the designer, the relationship with the city, the design of a space within the field of the exhibition branch, and the analysis of reference projects based on relevant themes. Based on a thorough analysis of the location in the city, the choice is made for the artefact/product and a specific user group. This allows the possibility to explore the full range of related interventions. Q2 BK-MI-210 During the second quarter, students design and build a fragment of the building on different scale levels (1:10/1:5 and sometimes even 1:1) based on the outcome of their space or display (or that of a fellow student) designed in Q1. This design project (BK7065 Spaces of Display 2) runs parallel to a Tools course (BK7067 Tools for Spaces of Display 2). The workshops and lectures in this quarter make a substantial contribution to the theme of the project.	
Study Goals	At the end of Q1 you will: have acquired knowledge, insights and skills in the domains of architectural and interior design, analysis, and presentation; have the skill to establish the connection between idea, function and material elaboration of a small scale design; have knowledge of different cultural and social conditions, and awareness of their implication and effects on architecture and the interior; be able to develop a consistent architectonic / interior design. This covers the materialization of the overall space as well as the detailing have developed abilities in the targeted analysis of design precedents and the effective representation and communication of findings, using various representation techniques as well as digital and physical models.	
Education Method	Excursion Design Tutorials Lectures Workshops	
Literature and Study Materials	Recommended literature: De Narratieve Ruimte ; Herman Kossmann Strategies of Display ; Julia Noordegraaf Exhibition Design - portfolio ; Philip Hughes Exhibition Design; Basic Interior Design 02 ; Pam Locker	
Assessment	Design project BK7064: final and midterm presentation with drawings/posters and models.	
Special Information	The excursion in the first week will be to a city in the Netherlands or Belgium. You will have to pay for your own travel expenses and entrance fees.	
Period of Education	Quarter 1	
Concept Schedule	Tuesday- and Friday mornings design tutorials Tuesday afternoon lectures and workshops Wednesday excursions 2-3 days	
Leerstoel	Interiors Buildings Cities	
Minimum number of participants	18 students	
Maximum number of participants	27 students	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

BK7066	Tools for Spaces of Display 1	5
Course Coordinator	Ir. L.M.M. de Wit	
Instructor	Ir. S. Pietsch	
Responsible for assignments	Ir. L.M.M. de Wit	
Contact Hours / Week x/x/x/x	X/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Required for	Bachelor students Architecture and Industrial Design (TU Delft and Eindhoven) Extern HBO: Design Academy Eindhoven Art academies; graduates in Architectural Design, interior architecture.	
Expected prior knowledge	Bachelor 1 to 4 or equivalent. or: Three years of HBO or advanced Bachelor in abovementioned educations. Recommended skills: Autocad, Illustrator, Sketchup or equivalent.	
Summary	The Chair of Interiors Buildings Cities is concerned with making buildings, in places, for people. It conceives of the city, at each scale, as a work of architecture and, hence, the responsibility of the architect. This developing discussion of the chair is contextualized through an annual theme. During this interiors minor, students design a (small) exhibition space with a fully detailed fragment of the building or interior. The minor consists of two quarters, each divided into two separate courses: The Design project and the Tools course. Please be aware that each quarter of this minor has a separate minor code. If you would like to follow both quarter 1 and 2, you should register for both minor codes in Osiris.	
Course Contents	The Tools Course (BK7066 Tools for Spaces of Display 1) focuses on specific skills that contribute to a coherent design project. Students learn how to work within the complex combination of these 'tools'. Various workshops and lectures focus on the position of the designer, the relationship with the city, the design of a space within the field of the exhibition branch, and the analysis of reference projects based on relevant themes. Based on a thorough analysis of the location in the city, the choice is made for the artifact/product and a specific user group. This allows the possibility to explore the full range of related interventions. During the workshops, specific knowledge of particular use to the design project will be gained from various discrete exercises. Experts will transfer their knowledge in a compact course in which students will be asked to make small studies in order to apply this knowledge directly. Important themes are location, the user and the artefact, quality of the perception of the interior and the materialization and detailing of space itself. We emphasize on the use of scale models as a means of research. This minor consists of 2 quarters (both of 15 ECs); each with separate content. For students of the Faculty of Architecture and the Built Environment TU Delft it is possible to follow either just quarter 1 or just quarter 2, or to follow both quarter 1 and 2. For all other students who are admissible to this minor, it is only possible to follow either quarter 1, or both quarter 1 and 2.	
Study Goals	- Has gained knowledge about the cultural and technical aspects of specific themes (for example: user research, materialization and detail) - Has learned to translate fresh knowledge into a design by doing a short practical study. - Is able to connect aspects like location, program, target group, functionality and temporariness to the atmospheric quality of building and space;	
Education Method	Lectures, Analysis, Intensive practical exercises with consults	
Assessment	Assignments, Lectures, Analysis and Workshops presented in a portfolio	
Period of Education	Quarter	
Leerstoel	Interiors Buildings Cities	
Minimum number of participants	18 students	
Maximum number of participants	27 students	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Bachelor Architecture, Urbanism and Building Sc.

BK-MI-210-23

BK7065	Spaces of Display 2	10
Course Coordinator	Ir. L.M.M. de Wit	
Instructor	Ir. S. Pietsch	
Responsible for assignments	Ir. L.M.M. de Wit	
Contact Hours / Week x/x/x/x	0/X/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor 1 to 4 or equivalent. or: Three years of HBO or advanced Bachelor in above mentioned educations. Recommended skills: Autocad, Illustrator, Sketchup or equivalent.	
Summary	The Chair of Interiors Buildings Cities is concerned with making buildings, in places, for people. It conceives of the city, at each scale, as a work of architecture and, hence, the responsibility of the architect. This developing discussion of the chair is contextualized through an annual theme. During this interiors minor, students design in Q1 a (small) exhibition space (BK7064) and in Q2 a fully detailed fragment of the building or the interior (BK7066). We emphasize on the use of scale models as means of research. Please be aware that each quarter of this minor has a separate minor code. If you would like to follow both quarter 1 and 2, you should register for both minor codes in Osiris.	
Course Contents	During the second quarter, students design and build a fragment of the building or the interior on different scale levels (1:10/1:5 and sometimes even 1:1) based on the outcome of their space or display (or that of a fellow student) designed in Q1. This design project (BK7065 Spaces of Display 2) runs parallel to a Tool course (BK7067 Tools for Spaces of Display 2). The workshops and lectures in this quarter make a substantial contribution to the theme of the project. Q1 BK-MI-209 During the first quarter, students design a small exhibition space for a carefully selected artifact and its users at a specific location in a city (BK7064 Spaces of Display 1). The Tools Course (BK7066 Tools for Spaces of Display 1) focuses on specific skills that contribute to a coherent design project. Students learn how to work within the complex combination of these 'tools'. Various workshops and lectures focus on the position of the designer, the relationship with the city, the design of a space within the field of the exhibition branch, and the analysis of reference projects based on relevant themes. Based on a thorough analysis of the location in the city, the choice is made for the artifact/product and a specific user group. This allows the possibility to explore the full range of related interventions. Q2 BK-MI-210 During the second quarter, students design and build a fragment of the building or the interior on different scale levels (1:10/1:5 and sometimes even 1:1) based on the outcome of their space or display (or that of a fellow student) designed in Q1. This design project (BK7065 Spaces of Display 2) runs parallel to a Tool course (BK7067 Tools for Spaces of Display 2). The workshops and lectures in this quarter make a substantial contribution to the theme of the project. This minor consists of 2 quarters (both of 15 ECs); each with separate content. For students of the Faculty of Architecture and the Built Environment TU Delft it is possible to follow either just quarter 1 or just quarter 2, or to follow both quarter 1 and 2. For all other students who are admissible to this minor, it is only possible to follow either quarter 1, or both quarter 1 and 2. Please be aware that each quarter has a separate minor code. If you would like to follow both quarter 1 and 2, you should register for both minor codes in Osiris.	
Study Goals	At the end of Q2 you will: have the skill to establish the connection between idea, function and material elaboration of a fragment of the building / interior; have acquired knowledge, insights and skills in the domains of furniture / interior design, analysis and presentation; have developed abilities in the targeted analysis of design precedents and the effective representation and communication of findings, using various representation techniques as well as digital and physical models; be able to develop a consistent building/interior fragment. This covers the materialization of the object itself as well as the detailed scale; the processing of technical and aesthetic aspects plays an essential part in this. .can determine the quality of the perception of the interior by the use of artificial light.	
Education Method	design tutorials	
Literature and Study Materials	Recommended literature: De Narratieve Ruimte ; Herman Kossmann Strategies of Display ; Julia Noordegraaf Exhibition Design - portfolio ; Philip Hughes Exhibition Design; Basic Interior Design 02 ; Pam Locker	
Assessment	Design project BK7065: final and midterm presentation with drawings/posters and models.	
Period of Education	Quarter 2	
Concept Schedule	Design Tutorials on Tuesdays- and Friday mornings	
Leerstoel	Interiors Buildings Cities	
Minimum number of participants	18 students	
Maximum number of participants	27 students	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

BK7067	Tools for Spaces of Display 2	5
Course Coordinator	Ir. L.M.M. de Wit	
Instructor	Ir. S. Pietsch	
Responsible for assignments	Ir. L.M.M. de Wit	
Contact Hours / Week x/x/x/x	0/X/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	Bachelor students Architecture and Industrial Design (TU Delft and Eindhoven) Extern HBO: Design Academy Eindhoven Art academies; graduates in Architectural Design, interior architecture	
Expected prior knowledge	Bachelor 1 to 4 or equivalent. or: Three years of HBO or advanced Bachelor in abovementioned educations. Recommended skills: Autocad, Illustrator, Sketchup or equivalent.	
Summary	The Chair of Interiors Buildings Cities is concerned with making buildings, in places, for people. It conceives of the city, at each scale, as a work of architecture and, hence, the responsibility of the architect. This developing discussion of the chair is contextualized through an annual theme. During this interiors minor, students design a (small) exhibition space with a fully detailed fragment of the building or the interior. The minor consists of two quarters, each divided into two separate courses: The Design project and the Tools course. Please be aware that each quarter of this minor has a separate minor code. If you would like to follow both quarter 1 and 2, you should register for both minor codes in Osiris.	
Course Contents	The Tools Course (BK7067 Tools for Spaces of Display 2) focuses on specific skills that contribute to a coherent design project; a fragment of the building or the interior. Students learn how to work within the complex combination of these 'tools'. Various workshops and lectures focus on f.i. the analysis of reference projects based on relevant themes, the position of the designer and the joiner, the design of a fragment of the building or a display element. During the workshops, specific knowledge of particular use to the design project will be gained from various discrete exercises. Experts will transfer their knowledge in a compact course in which students will be asked to make small studies in order to apply this knowledge directly. Important themes are, the user of the space and the designed fragment, displaying products, the materialization and detailing of the fragment itself. We emphasize the use of scale models as a means of research. This minor consists of 2 quarters (both of 15 ECs); each with separate content. For students of the Faculty of Architecture and the Built Environment TU Delft it is possible to follow either just quarter 1 or just quarter 2, or to follow both quarter 1 and 2. For all other students who are admissible to this minor, it is only possible to follow either quarter 1, or both quarter 1 and 2.	
Study Goals	The student: - Has gained knowledge about the cultural and technical aspects of specific themes (for example: user research, lighting and detailing) - Has learned to translate fresh knowledge into a design by doing a short practical study. - Is able to determine the quality of the perception of the interior by using artificial light. - Is able to connect aspects like program, user group, functionality, temporariness and the connection of atmospheric quality of the fragment in the space.	
Education Method	Lectures, Analysis, Intensive practical exercises with consults	
Assessment	Assignments Lectures, Analysis and Workshops presented in a portfolio	
Period of Education	Quarter	
Concept Schedule	lectures, workshops on Tuesday afternoons	
Leerstoel	Interiors Buildings Cities	
Minimum number of participants	18 students	
Maximum number of participants	27 students	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Bachelor Architecture, Urbanism and Building Sc.

BK-MI-2016-23

BK7215	Basic techniques for sustainable urban design	5
Course Coordinator	Ir. M.G.F. Overschie	
Responsible for assignments	Ir. M.G.F. Overschie	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	In a series of interactive lectures and workshops, present and future urban tasks will be presented and discussed, and urban design principles for sustainable urban plans will be practiced. Innovative realized plans will be introduced as examples. Sustainable urbanism themes such as climate change, the water challenge, heat stress, biodiversity, ecology, urban agriculture, air quality, energy, mobility, urban transport, waste and material flows, social and economic value, and process will be discussed. By field trips future proof sustainable plans will be illustrated. The edX MOOC Sustainable Urban Development will provide basic knowledge on global challenges and sustainable urbanism themes.	
Study Goals	In the urbanism professions, visualizations play an important role. They serve to communicate research results and planning proposals. But they are also used to develop ideas, in brainstorming and design. In this course, the use of visual means at each stage of the research and design process will be discussed. Students will work on visualisation assignments, such as: sketching to develop and share our ideas; mapping to understand the conditions, the history and to orient ourselves; and info graphs to explain an idea or to present statistics. After finishing this course, students: * Have the ability to recognize sustainable plans and sustainable measures (people, planet and profit) in urban plans. * Have knowledge of sustainable urbanism themes, e.g climate change effects and the spatial implications of adapting to climate change in sustainable urban plans. * Have basic skills to understand and use visual instruments, e.g. sketching, mapping and using icons.	
Education Method	Series of lectures, workshops and field trips (in and around Delft by bike), combined with self-study (edX MOOC Sustainable Urban Development).	
Assessment	* Take home exam on the content of the lectures, workshops, videos, and literature displayed on Brightspace. * Complete self-paced edX MOOC Sustainable Urban Development.	
Enrolment / Application	This course is part of the minor Sustainable Urbanism.	
Special Information	The maximum marking period of the take home exam is 15 work days.	
Period of Education	2nd quarter (re-take of the take home exam in 3th quarter)	
Leerstoel	Environmental Technology & Design	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

BK7225	City and Public Space: Sustainable Urban Design	10
Course Coordinator	Ir. M.G.F. Overschie	
Responsible for assignments	Ir. L.P.J. van den Burg	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	This part of the minor "Sustainable Urbanism: the Green Blue City" is integrated in the framework of the regular Bsc-3 design module ON3 in the environmental design variant. Course overview The programme consists of two courses of a total of 15 ECs. City and Public Space: Sustainable Urban Design (studio) (BK7225): 10 ECs Basic Techniques for Sustainable Urban Design: 5 ECs The task of the urban planner is to introduce and design sustainable urban interventions, creating future proof neighbourhoods within the framework of a green blue city. In the Sustainable Urban Design studio City and Public Space of this minor a sustainable urban design is the main subject. In this studio the minor students will develop design proposals, in interdisciplinary teams. Focus in the studio is on concrete sustainable design exercises. The course Basic Techniques for Sustainable Urban Design will make the minor students equipped and prepared to perform well in the design studio. They will apply theory and use their knowledge background in the group design process. In this minor you will learn: to recognize inner-city problems caused by climate change and social aspects; to recognize sustainable urban plans and sustainable measures in urban plans; use knowledge of climate change effects and the spatial implications of adapting to climate change in sustainable urban plans; to develop, integrate and evaluate spatial interventions for a future proof urban plan; to collaborate with students from other disciplines in order to improve own (basic) urban design skills, research skills and writing skills, and help improve these skills of others; to argue comprehensively for choices made in the design process, considering spatial, social and procedural circumstances; to give relevant feedback on content and presentation to peers, and accept and react on received feedback. For whom? The minor is intended for students from urban planning, regional planning, civil engineering, landscape architecture, technology policy and management, industrial design engineering, social sciences, and other studies related to the topic of climate change and the city. The minor is closed for architecture students from Delft University of Technology, however open to architecture exchange students. You must have interest and/or affection for design as may be expected from your respective expertise and bachelor programme. Furthermore, you have an interest in social and sustainable aspects and you are academically curious. This minor has an interdisciplinary character and thus will assemble students in groups to create a balanced variety of faculty backgrounds. For students from outside Delft University of Technology, Leiden University or Erasmus University a specific selection procedure applies. After applying as described on the minor website you will receive an email within three weeks in which you will be asked to fill in a questionnaire. After receiving your questionnaire, you will be informed whether you will be officially enrolled or not.	
Study Goals	General; the student - is able to make a design for a wide range of assignments, taking into account the relevant requirements and preconditions; - is able to make a design based on self-formulated principles, using a wide range of research techniques and is able to critically reflect on the quality of the design; - understands a wide range of basic technical requirements, sees them as an intrinsic part of designing and applies them as such. Specific; the student - is able to dissect the urban landscape into networks, allotment patterns, types of buildings and public spaces and is able to integrate these in a sustainable design, on different scales; - is able to control multiple scales simultaneously and in conjunction in the design. The student shows an understanding of the relationship between the main urban networks, allotment patterns and block typology. - can make urban design analyses based on design questions. The student can design in variants and relate these to the results of the analyses. The student is able to make calculations with regard to densities and take the results into account in the design. - can document the design process in urban design impressions, plan maps, profiles and models.	
Education Method	design studio	
Assessment	2 intermediate presentations in week, final presentation in week 9. One final grade is given on the basis of the quality of the final portfolio, deadline in week 10. If the course is graded as insufficient, the whole course has to be repeated at a later time or in the two-week summer school. The latter option is open only for students with a grade 5 of 5,5	
Period of Education	2	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Bachelor Architecture, Urbanism and Building Sc.

BK-MI-231-23 (Re)Imagining Port Cities: Understanding Space, Society and Culture

BK7940	(Re)imagining Port Cities: Understanding Space, Society and Culture Q1	15
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	Ir. E.J.G.C. van Dooren	
Responsible for assignments	Ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	Designing is imagining the future. In a time of climate change and multiple transitions, a "designerly" attitude is needed more than ever. At the same time, understanding past practices and sociocultural conditions is an essential basis for new interventions. In this minor, the phenomenon of port cities is studied from cultural, social and historical perspectives. Research studies of the city's past and present are conducted to learn about the city. And interventions are designed, ideas conceived and tested to make the city future-proof. To learn to think like a designer, the basic skills of the design process are made explicit as much as possible. In this quarter (BK 7940), the emphasis is on understanding the port city and understanding the port city in a larger context. In the follow-up quarter (BK 7941), the focus is on people in the city and a free choice item.	
Study Goals	Key objectives in the minor are: - learning a 'designerly' attitude - learning from the past - learning to imagine a sustainable and social future Specifically, this means learning - to study (analyze and reflect on) the historical, social and cultural context in port cities in relation to the present and the future, - experimenting (exploring and testing) with possible alternative future ideas, supported by knowledge, - take position as a designer in relation to the future of port cities by choosing a theme or quality, - working out different scale levels, such as street furniture, public space, the borders of water and land, a port area and/or a city vision.	
Education Method	lectures and tutorials in the (design) studio on the research and design studies	
Assessment	The assessment is based on the result of the research and design studies. The results are presented on posters with text and visualizations, such as images, photographs, diagrams and sketches	
Period of Education	quarter 1	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

BK7941	(Re)imagining Port Cities: Understanding Space, Society and Culture Q2	15
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	Ir. E.J.G.C. van Dooren	
Responsible for assignments	Ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	Designing is imagining the future. In a time of climate change and multiple transitions, a "designerly" attitude is needed more than ever. At the same time, understanding past practices and sociocultural conditions is an essential basis for new interventions. In this minor, the phenomenon of port cities is studied from cultural, social and historical perspectives. Research studies of the city's past and present are conducted to learn about the city. And interventions are designed, ideas conceived and tested to make the city future-proof. To learn to think like a designer, the basic skills of the design process are made explicit as much as possible. This quarter (BK 7941) focuses on people in the city and a free choice item. This quarter is a continuation of the first (BK 7940), which focuses on understanding the port city and understanding the port city in a larger context.	
Study Goals	Key objectives in the minor are: - learning a 'designerly' attitude - learning from the past - learning to imagine a sustainable and social future Specifically, this means learning - to study (analyze and reflect on) the historical, social and cultural context in port cities in relation to the present and the future, - experimenting (exploring and testing) with possible alternative future ideas, supported by knowledge, - take position as a designer in relation to the future of port cities by choosing a theme or quality, - working out different scale levels, such as street furniture, public space, the borders of water and land, a port area and/or a city vision.	
Education Method	lectures and tutorials in the (design) studio on the research and design studies	
Assessment	The assessment is based on the result of the research and design studies. The results are presented on posters with text and visualizations, such as images, photographs, diagrams and sketches	
Period of Education	quarter 2	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Bachelor Architecture, Urbanism and Building Sc.

BK-MI-235-23 Spatial Computing for Digital Twinning

BK7951	Spatial Decision Support Systems	5
Course Coordinator	Dr. A. Rafiee	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	The course provides a basis for spatial analysis in a large area. Students learn how to model and visualize the mutual impact of the environment and their design. Students also learn about solving complex spatial decision problems at a geographic scale, for which they apply various simulation and optimization approaches, among other things. The assignments are related to more sustainable development of the living environment, such as location allocation/planning of wind turbines and solar panels. The course will consist of a combination of lectures and hands-on labs. Students work in groups of 2-3 on an assignment around a complex spatial decision problem. The individual level of the students is assessed by means of the exam.	
Study Goals	After the course the student is able to: - Understand the complexity of spatial problems and the role of Spatial Decision Support Systems (SDSS) - Explain the concept, components and underlying technologies of SDSS - Apply Multicriteria Decision Analysis (MCDA) for spatial decisions - Use Multi-criteria Spatial Decision Support System (MC-SDSS) when making decisions about the (mutual) impact of a design and its environment - Compare design or decision alternatives according to the design environment effects and evaluation criteria	
Education Method	Lectures/labs 32 hours; 26 hours Assignments; 82 hours self-study. The lectures focus on Knowledge and Insight. The group projects focus on Applying the Knowledge and Insight. For the group projects, 2-3 students per group will work together.	
Assessment	Written exam and group assignments. The result of the written exam determines 50% of the final score and the group assignments 50%.	
Period of Education	Quarter 1	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

BK7952	3D Data Acquisition and Analysis	5
Course Coordinator	Dr. R.C. Lindenbergh	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	This course provides students with an overview and hands-on experience with 3D spatial data acquisition techniques and methodology for information extraction from 3D data, linked to urban applications. The course consists of a combination of lectures and peer-to-peer presentations by students. Students work in groups of three on an integrating assignment around a relevant topic about acquisition, information extraction and application. The individual level of the students is assessed by means of a short written exam	
Study Goals	After completing the 3D Data Collection and Analysis course, students will be able to: - Explain the principles and properties of 3D point cloud acquisition techniques, including in particular photogrammetry, laser scanning and multibeam echosounding - Draw conclusions about the accuracy and precision obtained with different measurement setups and processing techniques - Extract information from point clouds in a suitable computing environment - Discuss the advantages and disadvantages of different techniques for extracting information from point clouds	
Education Method	- Lectures/labs including peer presentations: 40 hours - 40 hours assignment; - 60 hours self-study: The lectures and small exercises focus on knowledge and insight and require active student participation The group project (i.e. big assignment) focuses on learning by doing and inquiry based knowledge acquisition with 2-3 students per group	
Assessment	Big assignments and a small written exam. The result of the big assignment determines 60% of the final score and the small written exam 40%.	
Period of Education	Quarter 1	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

BK7953	Open Urban Data Governance	5
Course Coordinator	Dr.ir. B. van Loenen	
Course Coordinator	mr.dr. H.D. Ploeger	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Required for	This course is part of the minor Spatial Computing for Digital Twinning.	
Course Contents	This course aims to provide a general insight into the relevant legislation and regulations as well as organizational aspects involved in the collection, processing, provision and use of individual datasets in the built environment. In small groups, the students will answer a specific social sustainability related question in the context of the built environment with open urban (geo) data.	
Study Goals	After taking this course, students are expected to be able to: - Apply relevant national and international legislation and policy (Open Government Act, Land Registry Act) regarding the collection, processing, provision and use of individual datasets - Apply organizational concepts (open data, FAIR data, black data, fake data, master data, data management) regarding the collection, processing, provision and use of individual datasets - Provide insight into data user needs and report on them - Recognize opportunities and limitations in the use of the data from a legal and/or organizational perspective - Analyse a specific case with regard to the design and use of the city in order to design a framework for optimal use of the data.	
Education Method	Workshops (14 hours)/ group work (86 hours)/self study (40 hours)	
Assessment	The course has two summative assessment methods: 1. Writing a paper about their case study (group assignment) (max. 5000 words, references included). 2. Individual oral exam (open book). The final mark will be based on the following: paper: 80%, individual exam: 20%. Students need to get at least a sufficient grade (5,75) for both the exam and the group assignment in order to obtain the overall grade for the course. Failing the group assignment will result in a new group assignment.	
Period of Education	1	
Minimum number of participants	15	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Bachelor Architecture, Urbanism and Building Sc.

BK-MI-236-23 Timeless Typologies - "Remodelling Architecture"

BK7942	Culture & Understanding	5
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	5 hours per week	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	The minor Timeless Typologies will focus on studying and modelling historical architectural precedents, covering a known oeuvre or style. The chosen theme will be represented by a series of case study projects, regarded as well-respected in the architectural field. The main goal is for students to understand which historical, spatial and technical values of the architectural precedents can be related to present architectural practice. The courses Culture & Understanding (History) and Design & Remodelling (Form Studies) are both integrated in this minor.	
Study Goals	At the end of the minor you: have acquired knowledge, insights, and skills in the domains of architectural research, design, analysis, and presentation; have developed abilities in the targeted analysis of design precedents and the effective representation and communication of findings, using various representation techniques, with a focus on physical models regarding Design & Remodelling have the ability to work individually and in a team, carrying out research, documenting results, visualizing proposals and bringing these across to a professional audience.	
Education Method	Lectures, workshops, feedback sessions, physical modelling : all in design studio format.	
Assessment	For Culture & Understanding: Essay, based on first impressions and literature research. Research report. Written dossier, relating the project to essay in a reflective matter. If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Period of Education	Quarter	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

BK7943	Design & Remodelling	10
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	The minor Timeless Typologies will focus on studying and modelling historical architectural precedents, covering a known oeuvre or style. The chosen theme will be represented by a series of case study projects, regarded as well-respected in the architectural field. The main goal is for students to understand which historical, spatial and technical values of the architectural precedents can be related to present architectural practice. The courses Culture & Understanding (History) and Design & Remodelling (Form Studies) are both integrated in this minor.	
Study Goals	At the end of the minor you: have acquired knowledge, insights, and skills in the domains of architectural research, design, analysis, and presentation; have developed abilities in the targeted analysis of design precedents and the effective representation and communication of findings, using various representation techniques, with a focus on physical models regarding Design & Remodelling have the ability to work individually and in a team, carrying out research, documenting results, visualizing proposals and bringing these across to a professional audience.	
Education Method	Lectures, workshops, feedback sessions, physical modelling : all in design studio format.	
Assessment	For Design & Remodelling Analysis in sketch models, drawings and written group manifest. Presentation: Models, posters, oral in exhibition setting. If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Period of Education	2e Quarter	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Bachelor Architecture, Urbanism and Building Sc.

BK-MI-241-23 Heritage and Design

BK7550	Landscape and Transition	5
Course Coordinator	Dr.ir. G.A. Verschuure-Stuip	
Responsible for assignments	Dr.ir. G.A. Verschuure-Stuip	
Co-responsible for assignments	Dr. I. Nevzgodin	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	Continuity and transformation are the keywords in our urban environment. The transformation of our built environment is of all ages and is currently one of the most urgent and challenging tasks in our society. In the design studio Landscape and Transition, you are asked to make a concept and design for a part of the green urban heritage in the city. The basis for an intervention is historic spatial analysis of the site, knowledge of the landscape architectural, urban and architectural history as well as current issues and problems in the urban public space. The challenge is to preserve monumental values and existing qualities as part of the sense of the place and combine this with new functions for up-to-date use. Studies of various methods of transformation of green heritage is part of the framework and will be studied and presented in reference projects. During the course you will familiarise yourself with design skills, knowledge of city and architectural development of Dutch cities and landscapes, analysis methods and various theoretic approaches, as well as in cultural, societal and philosophical aspects. You will have the opportunity to gain knowledge in and to practise with aspects such as historical identity, value proposition and spatial intervention with objects and structures of cultural heritage in a historic environment.	
Study Goals	You will practise the interaction between history and design. After this course a student is able to successfully: - determine and draw the historic characteristics and current spatial qualities of specific cultural landscapes, green monumental heritage in word and drawings - Connect the characteristics of the historic urban landscape with new transformations and interventions. - make a simple landscape architectural plan for an intervention in a cultural landscape or urban green heritage site - reflect on the connection between preservation of the historic character, your plan and new interventions.	
Education Method	Design studio Landscape and Transition In the design studio, you are asked to produce a spatial design of a part of the site and work on specific details. To get to this design you will discuss and use various theories and methods in current and historic analysis as well as various design skills (architectural design, sketching, mood boards, collage, models, etc.).	
Assessment	The design studio will round up with a final presentation of the analysis and design as well as a presentation of reference projects. The course as well as the minor is successfully completed if you complete all study parts with grade 6 or higher.	
Special Information	The course Landscape and Transition is part of the minor City and Landscape; heritage and transformation in the first quarter. It is preliminary for the minor Design: Architecture and Heritage in the second quarter. The programme of the entire minor consists of one lecture series and two studio assignments, they account for a total of 15 study points whereby each part is worth 5 study points. BK7551 History of Dutch cities and Landscapes (lecture series) About the development of the Dutch city and the Dutch landscape. BK7555 Research Studio City and Transformation (studio assignment) Focuses on the analysis and assessment of the city and its buildings. The analysis must ultimately result in a cultural-historic value proposition. BK7550 Landscape and Transition (studio assignments) For the design studio Landscape and Intervention, a design must be made of the intervention of a public space, in a historic urban or landscape environment. The contact hours for the studio assignment is 8 hours per week. The contact hours for the lecture series can be maximum 8 hours per week. Included self-study the student is supposed to spend (15x28=) 420 hours on the whole course.	
Period of Education	Quarter 1	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

BK7551	History of Dutch Cities and Landscapes	5
Course Coordinator	Dr. R.J. Rutte	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Responsible for assignments	Dr. R.J. Rutte	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	Recommended: BK 7554 History of Dutch architecture and art	
Course Contents	The Netherlands is the most densely urbanized country in Europe. Its crowded landscape of greater and smaller, older and younger towns was formed in the course of a millennium. What were the roots of this urban landscape, and how did it develop?	
Study Goals	After seven weeks every student knows the main characteristics and periods of urbanization, landscape transformation, and architecture in the Netherlands from the tenth to the twenty-first century, and is able to place these developments in their historical, economic, political and demographical context. The student understands that the form of spatial patterns and buildings depends on specific historical processes and can answer the question: Why do today's Dutch towns and cities, man-made landscapes and architecture look the way they do?	
Education Method	Lectures Field trips	
Literature and Study Materials	R. Rutte, The Making of the Netherlands. Landscapes, Cities and Architecture, Bussum 2023 or R. Rutte, Historische atlas van Nederland. Hoe ons land in 2000 jaar is ingericht, Bussum 2023.	
Assessment	written examination	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

BK7552	Heritage: Theory and Practice	5
Course Coordinator	Dr. I. Nevzgodin	
Instructor	A.C. de Ridder	
Instructor	Dr. I. Nevzgodin	
Responsible for assignments	A.C. de Ridder	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Continuity and change are the keywords for this minor. The transformation of our built environment is of all ages and is currently an urgent design task. In this minor you will learn to develop concepts for the transformation of buildings, urban areas and public spaces. The basis for these interventions is a historic analysis of the site and knowledge of architectural history. The challenge is to preserve monumental values, existing qualities and the sense of place while giving way for an intervention that allows for a new function and prepares a site for an up-to-date use. Studies of heritage and other subjects form an important framework. You will immerse yourself in the issue of cultural-historic continuity and urban identity. In this minor the particular focus lies on how we currently handle cultural heritage and necessary urban renewal with reference to changing function requirements. To approach these complex tasks, you will be challenged to familiarise yourself with the specific design skills and knowledge of city and architectural development of Dutch cities. In addition, you will delve into analysis methods and various theoretic approaches, as well as in cultural, societal and philosophical aspects. You will have the opportunity to gain knowledge in and to practise with aspects such as historical identity, value assessment and spatial intervention with objects and structures of cultural heritage in a historic environment. History is used as inspiration for the design. This minor is intended for all students of TU Delft, Erasmus University and University of Leiden. As both students of architecture and other disciplines participate, the designs will be developed and presented at a conceptual level, for which it is not necessary to have all the knowledge and skills required for a conventional architectural design. We aim to work in trans-disciplinary teams, where all team members can bring in their creativity, visions and ambitions on an equal base. External academic: art history, architectural history, archaeology, landscape architecture, planning. Level: end second year BSc.	
Study Goals	You will learn about the concept of cultural heritage, and the interaction between history and design. At the end of the minor you will have enhanced/improved your: knowledge about different concepts and scale levels of heritage and preservation, from academic research to architectural practice. knowledge about relevant processes in urban history and development. knowledge about approaches and work methods concerning appreciation, re-purpose and redevelopment of historically valuable objects within an old town. understanding of Architectural Heritage and relevant terminology (e.g. Values of Heritage, Intangible Cultural Heritage, Integrity, Reversibility, and Authenticity, etc.). ability to analyse heritage buildings and ensembles with regards to the urban context and its historic development, the functionality, the materials used and the technical development. ability to justify your point of view in discussions about the valuation of heritage on different scales and about the merging of old and new in case of an intervention.	
Education Method	Lectures series and one workshop.	
Literature and Study Materials	Literature and Study Materials will be available two weeks prior to the start of the course in Brightspace.	
Assessment	The lecture series are rounded off with an written exam. The minor is successfully completed if you complete all study parts with grade 6 or higher. All grades of 5.5 or lower are eligible for a written retake in the week 3.5.	
Special Information	The maximum marking period is 10 work days. The programme of the Minor Heritage & Design consists of three lecture series and three studio assignments, they account for a total of 30 study points whereby each part is worth 5 study points. BK7551 History of Dutch Cities and Landscapes (lecture series) The development of the Dutch city and the Dutch landscape. BK7555 City and Transformation (studio assignments) The first research studio City and Transformation focusses on the analysis of the city and its buildings. The analysis must ultimately result in a cultural-historic value proposition. This value proposition is used as boundary requirement for a design to be created in the subsequent studios. BK7550 Landscape and Transition (studio assignments) For the design studio Landscape and Transition, a design must be made of the intervention of a public space, in a historic urban or landscape environment. BK7554 History of Dutch architecture and art (lecture series) In this series of lectures and excursions the stylistic, typological and iconographic developments of Dutch art and architecture from the tenth to the twenty-first century will be studied in close detail. BK7552 Heritage: Theory and Practice (lecture series) Different concepts and scale levels of heritage and preservation, from academic research to architectural practice. BK7553 Architecture and Re-use (studio assignments) For the studio assignment you will work on the transformation of a historic building to which a new function is added. The contact hours for the lecture series are ca. 4 hour per week starting from week 2.1 and ending in week 2.6. Included self-study the student is supposed to spend (15x28=) 420 hour on the whole course.	
Period of Education	The minor Heritage & Design is offered in the first semester. The different subjects of the program will be divided over the semester.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

BK7553	Architecture and Re-use	5
Course Coordinator	Dr. I. Nevzgodin	
Course Coordinator	Ir. A.C. de Ridder	
Responsible for assignments	Ir. A.C. de Ridder	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	Continuity and change are the keywords for this minor. The transformation of our built environment is of all ages and is currently an urgent design task. In this minor you will learn to develop concepts for the transformation of buildings, urban areas and public spaces. The basis for these interventions is a historic analysis of the site and knowledge of architectural history. The challenge is to preserve monumental values, existing qualities and the sense of place while giving way for an intervention that allows for a new function and prepares a site for an up-to-date use. Studies of heritage and other subjects form an important framework. You will immerse yourself in the issue of cultural-historic continuity and urban identity. In this minor the particular focus lies on how we currently handle cultural heritage and necessary urban renewal with reference to changing function requirements. Related to the urban renewal we also focus on the use of public areas as well as the interior spaces of historic buildings in an old city centre. To approach these complex tasks, you will be challenged to familiarise yourself with the specific design skills and knowledge of city and architectural development of Dutch cities. In addition, you will delve into analysis methods and various theoretic approaches, as well as in cultural, societal and philosophical aspects. You will have the opportunity to gain knowledge in and to practise with aspects such as historical identity, value assessment and spatial intervention with objects and structures of cultural heritage in a historic environment. History is used as inspiration for the design. This minor is intended for all students of TU Delft, Erasmus University and University of Leiden. As both students of architecture and other disciplines participate, the designs will be developed and presented at a conceptual level, for which it is not necessary to have all the knowledge and skills required for a conventional architectural design. We aim to work in trans-disciplinary teams, where all team members can bring in their creativity, visions and ambitions on an equal base. External academic: art history, architectural history, archaeology, landscape architecture, planning. Level: end second year BSc.	
Study Goals	You will practice the interaction between history and design. At the end of the minor you will have enhanced/improved your: knowledge about relevant processes in urban history and development; knowledge about cultural heritage in the form of urban projects and building structures; knowledge about approaches and work methods concerning appreciation, repurpose and redevelopment of historically valuable objects within an old town; ability to analyse complex heritage buildings and ensembles with regards to the urban context and its historic development, the functionality, the materials used and the technical development; understand and deal with complexity of the monument; ability to make architectural and spatial design choices on different scales, based on an analysis and evaluation of current historic information; ability to put historic research and contextual analysis in functional relation with a design task; ability to justify your point of view in discussions about the valuation of heritage on different scales and about the merging of old and new in case of an intervention.	
Education Method	One introduction lecture, the lectures from parallel course BK7552, and studio assignments. In the studios you will work on your own design proposal, using a variety of design skills (architectural design, sketching, mood boards, collage, models, etc.).	
Literature and Study Materials	Literature and Study Materials will be available two weeks prior to the start of the course in Brightspace.	
Assessment	The studio design assignments are rounded off with the presentation of design proposals for interventions in an historical building. The minor is successfully completed if you complete all study parts with grade 6 or higher. All grades of 5.5 or lower are eligible for repair in two weeks after the first group presentation.	
Special Information	The maximum marking period is 10 work days. The programme of the Minor Heritage & Design consists of three lecture series and three studio assignments, they account for a total of 30 study points whereby each part is worth 5 study points. BK7551 History of Dutch Cities and Landscapes (lecture series) The development of the Dutch city and the Dutch landscape. BK7555 City and Transformation (studio assignments) The first research studio City and Transformation focusses on the analysis of the city and its buildings. The analysis must ultimately result in a cultural-historic value proposition. This value proposition is used as boundary requirement for a design to be created in the subsequent studios. BK7550 Landscape and Transition (studio assignments) For the design studio Landscape and Transition, a design must be made of the intervention of a public space, in a historic urban or landscape environment. BK7554 History of Dutch architecture and art (lecture series) In this series of lectures and excursions the stylistic, typological and iconographic developments of Dutch art and architecture from the tenth to the twenty-first century will be studied in close detail. BK7552 Heritage: Theory and Practice (lecture series) Different concepts and scale levels of heritage and preservation, from academic research to architectural practice. BK7553 Architecture and Re-use (studio assignments) For the studio assignment you will work on the transformation of a historic building to which a new function is added. The contact hours for the studio assignments are 6 hours per week starting from week 2.3 and ending in week 2.8. The contact hours for the lecture series is ca. 8 hours a week. Included self-study the student is supposed to spend (15x28=) 420 hours on the whole course.	
Period of Education	The minor Heritage & Design is offered in the first semester. The different subjects of the program will be divided over the semester.	

Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).
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BK7554	History of Dutch architecture and art	5
Course Coordinator	mr.dr. E. Korthals Altes	
Responsible for assignments	mr.dr. E. Korthals Altes	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	In this series of lectures and excursions the stylistic, typological and iconographic developments of Dutch art from the fifteenth to the twenty-first century will be studied in close detail. It focuses on the representation of landscape, cities and architecture.	
Study Goals	After nine weeks every student knows the main characteristics and periods of Dutch art in the Netherlands from the fifteenth to the twenty-first century, and she/he is able to place these developments in their cultural, economic and political context. She/he understands that form and function of art works depend on specific historical processes and can answer questions such as: How does the visual representation of the architecture of the past influence the way we perceive the built environment today?	
Education Method	Lectures and excursions	
Literature and Study Materials	Reinout Rutte, The Making of the Netherlands. Landscape, Cities and Architecture, Bussum 2023. Additional literature will be provided during the course.	
Prerequisites	Advanced Bachelor (HBO 3 years WO 2 years).	
Assessment	Written examination	
Exam Hours	2	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 1	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

BK7555	City and Transformation	5
Course Coordinator	Dr. I. Nevzgodin	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	Continuity and change are the keywords for this minor. The transformation of our built environment is of all ages and is currently an urgent design task. In this Research Studio "City and Transformation" you will learn to understand and to develop concepts for the transformation of urban areas and public spaces. The basis for these interventions is a historic analysis of the site and knowledge of architectural history. The challenge is to preserve monumental values, existing qualities and the sense of place while giving way for an intervention that allows for a new function and prepares a site for an up-to-date use. Studies of heritage and other subjects form an important framework. You will immerse yourself in the issue of cultural-historic continuity and urban identity. In this minor the particular focus lies on how we currently handle cultural heritage and necessary urban renewal with reference to changing function requirements. Related to the urban renewal we focus on the use of public areas. To approach these complex tasks, you will be challenged to familiarise yourself with the specific analytical skills and knowledge of city and architectural development of Dutch cities. In addition, you will delve into analysis methods and various theoretic approaches, as well as in cultural, societal and philosophical aspects. You will have the opportunity to gain knowledge in and to practise with aspects such as historical identity, value assessment and spatial intervention with objects and structures of cultural heritage in a historic environment. This minor is intended for all students of TU Delft, Erasmus University and University of Leiden. As both students of architecture and other disciplines participate, the analyses will be developed and presented at a conceptual level, for which it is not necessary to have all the knowledge and skills required for a conventional architectural design. We aim to work in trans-disciplinary teams, where all team members can bring in their creativity, visions and ambitions on an equal base. External academic: art history, architectural history, archaeology, landscape architecture, planning. Level: end second year BSc.	
Study Goals	You will practice the interaction between history and design. At the end of this course you will have enhanced/improved your: knowledge about cultural heritage in the form of urban projects and structures; knowledge about relevant processes in urban history and development; to use professional terminology; knowledge about approaches and work methods concerning appreciation, repurpose and redevelopment of historically valuable objects within an old town; ability to analyse heritage buildings and ensembles with regards to the urban context and its historic development, the functionality, the materials used and the technical development; to apply different tools of spatial analysis and ability to explain your choice of them; ability to put historic research and contextual analysis in functional relation with a design task; ability to justify your point of view in discussions about the valuation of heritage on different scales and about the merging of old and new in case of an intervention.	
Education Method	Lectures, excursions and studio assignments. In the studios, you will work on the research in small groups. For the urban analysis assignments, we will work in a trans-disciplinary manner, using a variety of design skills (architectural design, sketching, mood boards, collage, models, etc.).	
Literature and Study Materials	Literature and Study Materials will be available two weeks prior to the start of the course in Brightspace.	
Assessment	The studio assignments are rounded off with the presentation of the analysis reports. The minor is successfully completed if you complete all study parts with grade 6 or higher. All grades of 5.5 or lower are eligible for repair in two weeks after the first group presentation.	
Special Information	The maximum marking period is 10 work days. The programme of the Minor Heritage & Design consists of three lecture series and three studio assignments, they account for a total of 30 study points whereby each part is worth 5 study points. BK7551 History of Dutch Cities and Landscapes (lecture series) The development of the Dutch city and the Dutch landscape. BK7555 City and Transformation (studio assignments) The first research studio City and Transformation focusses on the analysis of the city and its buildings. The analysis must ultimately result in a cultural-historic value proposition. This value proposition is used as boundary requirement for a design to be created in the subsequent studios. BK7550 Landscape and Transition (studio assignments) For the design studio Landscape and Transition, a design must be made of the intervention of a public space, in a historic urban or landscape environment. BK7554 History of Dutch architecture and art (lecture series) In this series of lectures and excursions the stylistic, typological and iconographic developments of Dutch art and architecture from the tenth to the twenty-first century will be studied in close detail. BK7552 Heritage: Theory and Practice (lecture series) Different concepts and scale levels of heritage and preservation, from academic research to architectural practice. BK7553 Architecture and Re-use (studio assignments) For the studio assignment you will work on the transformation of a historic building to which a new function is added. The contact hours for the studio assignment are 6 hour per week starting from week 1.1 and ending in week 1.6. The contact hours for the lecture series is ca. 4 hour per week. Included self-study the student is supposed to spend (15x28=) 420 hour on the whole course.	
Period of Education	The minor Heritage & Design is offered in the first semester. The different subjects of the program will be divided over the semester.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Bachelor Architecture, Urbanism and Building Sc.

Bachelor 6

BK6AC3	Academic Skills 3 Design Reflection	5
Course Coordinator	J. Gosseye	
Course Coordinator	Dr. J.L. Heintz	
Instructor	Dr. J.L. Heintz	
Responsible for assignments	Dr. J.L. Heintz	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	Dutch	

BK6MA3	Society, Process and Practice 3 Management, Planning and (Re)Development	5
Course Coordinator	Ir. E.H.M. Geurts	
Course Coordinator	Ir. H.W. de Wolff	
Responsible for assignments	Ir. H.W. de Wolff	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3 5	
Course Language	Dutch	

BK6ON5	Design 5 Area (Re)Development in the Metropolitan Landscape	10
Course Coordinator	Dr.ir. R.M. Rooij	
Course Coordinator	Ing. P. de Jong	
Instructor	Ing. P. de Jong	
Responsible for assignments	Ing. P. de Jong	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	Dutch	
Course Contents	URBAN AREA STRATEGY is teamwork. For the redevelopment of an existing multifunctional urban area an urban area strategy will be developed. This is the result of teamwork, in which a multi-actor decision process is simulated as a management game (roleplay). In this process an appraisal of interests, goals and means of stakeholders is conducted within the context of economical and societal developments. REDEVELOPMENT ASSIGNMENT The assignment for area development includes an existing urban area in which the current utility does not fulfil the spatial and economic strength. The area development assignment aims at an optimal positioning of the existing and/or desirable functions in the context of the current and future market demand. Part of this approach is a proper urban placement. Within the specifications of the redevelopment assignment the right balance between preservation, improvement and renewal is searched for. This includes existing building, urban elements and aboveground as well as subterranean infrastructure. MANAGEMENT GAME The content of the redevelopment assignment and possible solution, crystallised in an area strategy) is arranged as a team, in the structure of a management game (roleplay). A multidisciplinary taskforce working on a urban strategy is simulated. This implies the issue is approached by every team from different professional perspectives. Tuning of sub solutions is an important aspect of the task. Expert tutors offer the necessary tools, enabling the perception of spatial, functional and financial consequences of various plan variants. ELABORATION OF URBAN STRATEGY The urban strategy delivers the limiting conditions for further elaboration of buildings, public space and other parts of the area.	
Study Goals	The student is able in a team to integrate programmatic, functional, spatial, technical, operational en economical aspects in an urban strategy. The student is able to connect different scales, from strategic project level to regional scales, in both spatial and programmatic perspective. The student is part of a team and is able to balance interests of the individual role and the team interest in an iterative design process. The student conducts role and team research in order to elaborate the practical context (program, urban vision, position of stakeholders) and design educated alternatives. The student is able to manage as such in a team. The student is able to visualise and present in a team (group products) and validate in the role report the process and design choices.	
Education Method	MANAGEMENT GAME Team assistance (group tutor) and expert assistance (role tutor). Students are working in teams (9-10) in a role play for an urban redevelopment strategy on a given location (area). Students are guided by a group tutor in their preparations of the strategy. Simultaneously students are receiving additional education on a specific role, guided by the role tutor, together with all other students with the same role. WORKSHOPS Through a series of workshops relevant additional skills beyond the roles are trained.	
Assessment	ASSESSMENT The student is assessed on two basic results: [1] academic competences to be measured during the expert meetings and the role report. [2] contribution to the team process - to be measured during the group meetings, presentations and contribution to the products. Results are an individual role report (50% of the grade) and several group products (50% of the grade): poster and a comprehensive team presentation. Through peer review (students) and expert review additional feedback is given to students and tutors. Note both partial grades should be 5 or above. The quality of the products are also taken into account for the process grade and relevant for both reviews. ATTENDANCE The process grade is based on presentations, the team urban redevelopment strategy, attendance, individual response and reflection. Because non-attendance hampers proper judgement and disturbs the team process, attendance during the course is essential. RESIT The process grade, in case of insufficient result, could be adjusted through a specific, additional task. The role grade can be improved in the dedicated period.	
Special Information	The maximum time for grading 15 workdays. See for additional information https://citydealmanagementgame.nl/	
Period of Education	Quarter	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

BK6ON6	Design 6 Building andTechnology	10
Course Coordinator	Ir. C.N. van Leest	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	Dutch	

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